

Multimodal Foundational Models for BioMedical Imaging



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Tutorial plan

1. Introduction to Foundation Models -- 8.00 to 8:40 AM (Tanveer)

- Foundation models - Evolution, definition
- Transformer architecture
- Vision transformers
- Representation learning

2. Contrastive Vision-Language Models for Medical Imaging -- 8:40 to 9:20 AM (Ismail)

- Embedding prior knowledge
- Zero-shot and few-shot inference
- Open-vocabulary segmentation

3. Fine-tuning and adaptation of foundation models - 09:20 to 10:00 AM (Ismail)

- Parameter-efficient fine-tuning (PEFT): Prompt-learning, adapters, low-rank approximations
- Unsupervised fine-tuning and transductive learning
- **Coffee break: 10.00 to 10:30 AM**

Tutorial plan & a few introductory words

4. VLM for image generation -- 10:30 to 10:50 AM -- (Tanveer)

- Diffusion models
- Synthetic medical image generation

5. Multimodal LLMs for text generation -- 10:50 to 11:10 AM -- (Tanveer)

- Llava-style models
- Retrieval-augmented generation
- Inference scaling

6. Medical Multimodal LLMs -- 11:10AM to 12:20 PM -- (Yunsoo)

- MLLMs for radiology
- MLLMs for pathology
- Evaluating report generation for medical images

Closing -- 12:20PM to 12:30 PM -- Tanveer

- Summary and conclusions

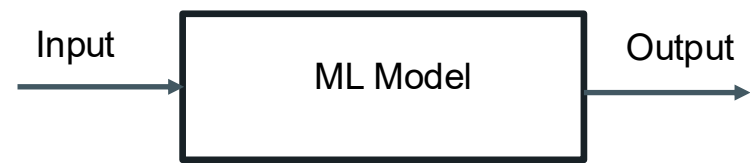
Notes

- Slides on self-supervised learning were derived from a lecture on that topic as part of the BMDS271 course at Stanford by Akshay Chaudhari
- Slides on foundation models were adapted from a lecture on that topic by Tanveer Syeda-Mahmood as part of the same course
- Due to animations, some slides may not render correctly. Our apologies.

Part 1

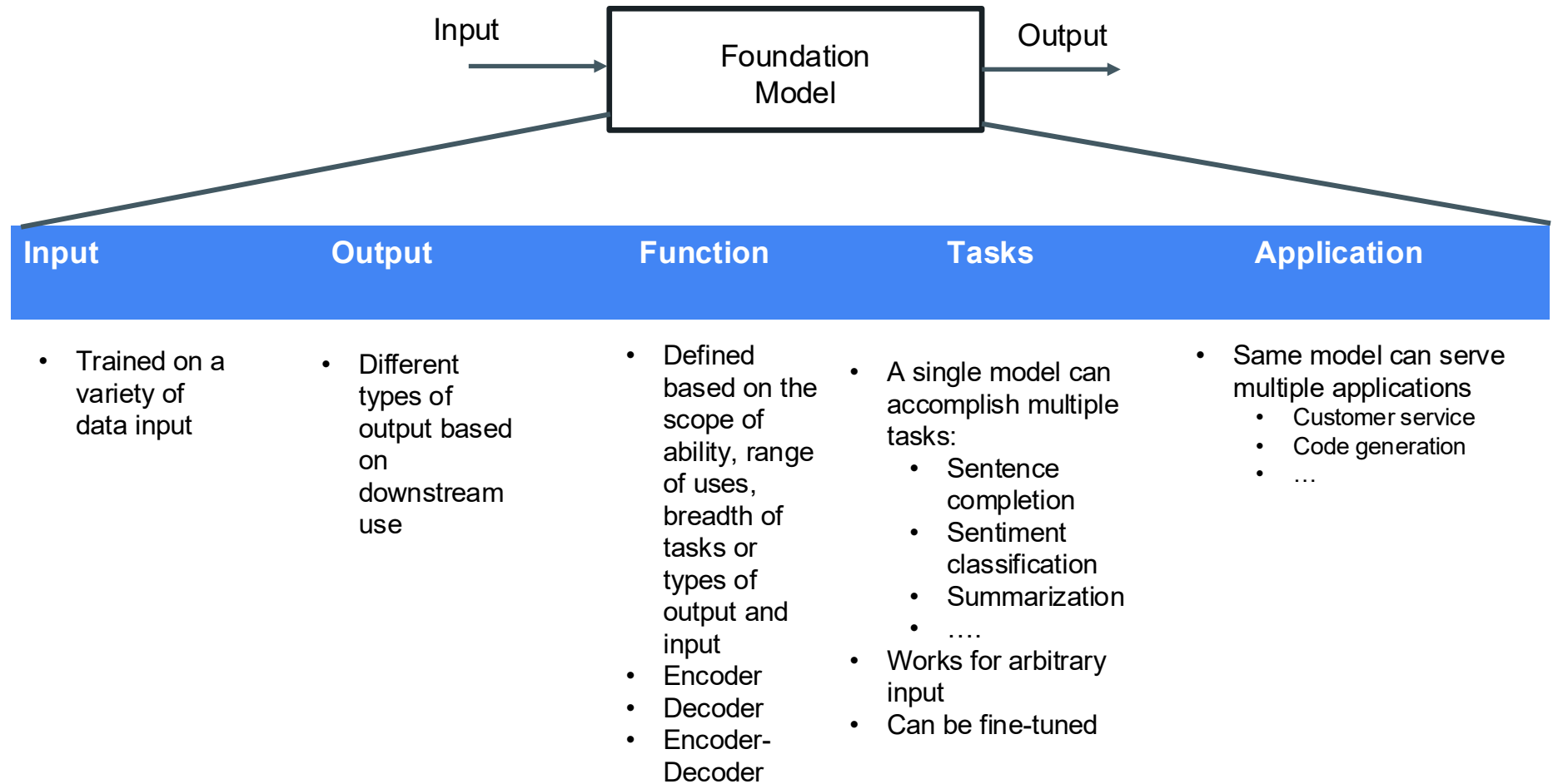
Introduction to Foundation Models

ML Models



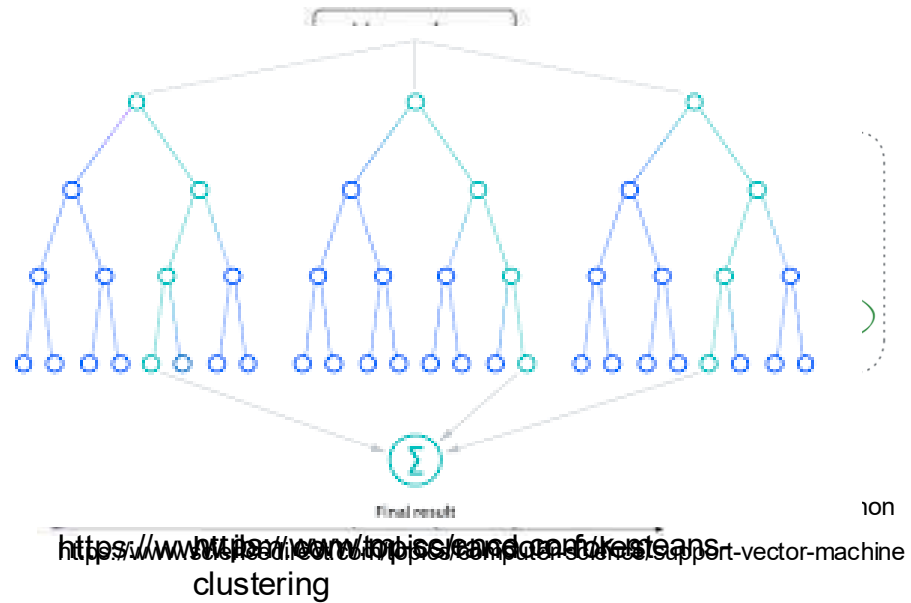
Input	Output	Function	Tasks	Application
• Structured • Time series • Text • Image • Audio • Video • Table • Domain-specific • Multi-D • Multimodal	• Label • Regression • Text • Image • Audio • Video • Table • Domain-specific	• Clustering • Classification • Prediction • Regression • Synthesis	• Recognition • Detection • Segmentation • Captioning • Image/text/synthesis • Audio/Video/multimodal synthesis • Question answering • Text generation • Autocompletion • Translation • Summarization • Sentiment analysis • Navigation • Search/IR • Recommendation •	• Customer service • Social Media • Marketing • Code generation • Automated reporting • Domain-specific

Foundation Model



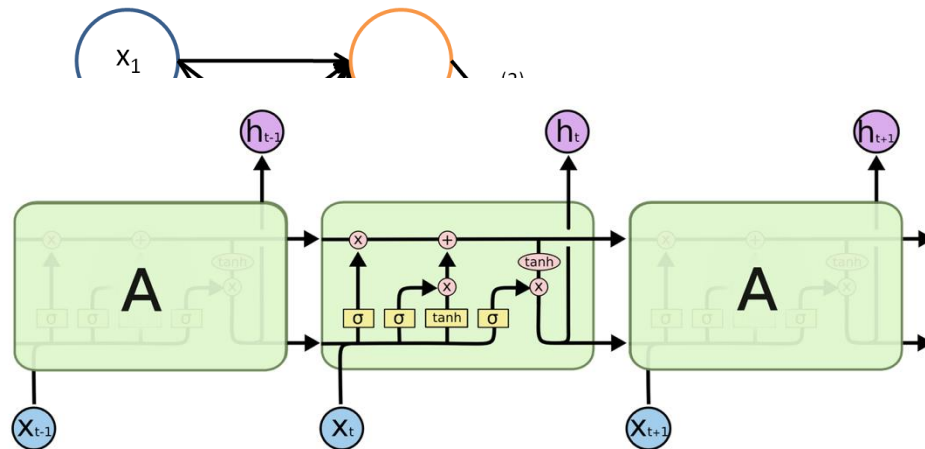
Are these Foundational Models?

- Decision Trees
- Clustering
- Support Vector Machines
- Random Forests



Are these Foundational Models?

- Multi-layer perceptron
- RNN
- LSTM

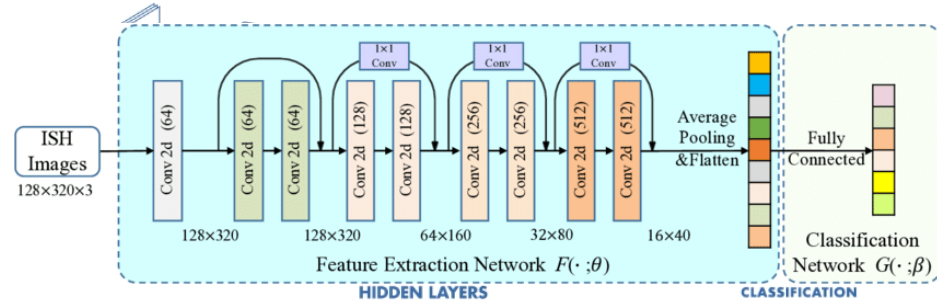


The repeating module in an LSTM contains four interacting layers.

<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>
<http://deeplearning.stanford.edu/tutorial/supervised/MultiLayerNeuralNetworks/>

Are these Foundational Models?

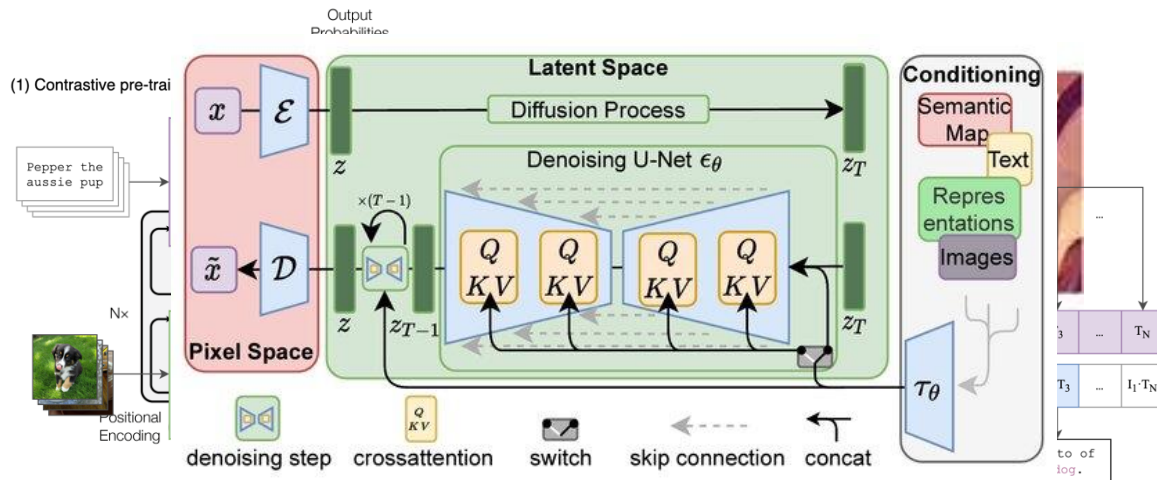
- Convolutional neural networks
- ResNet
- DNN



<https://www.linkedin.com/pulse/deep-residual-networks-resnet-ayoub-krouane>
<https://towardsdatascience.com/convolutional-neural-networks-explained-9cc5188c4939>

Are these Foundational Models?

- Auto-encoders
- GANs
- Transformers
- GPT
- CLIP
- Diffusion models



https://openaccess.thecvf.com/content/ICCV2022/papers/Rombach_High-Resolution_Image_Synthesis_via_Latent_Diffusion_Models_CVPR_2022_paper.pdf

<https://arxiv.org/abs/1706.03762>

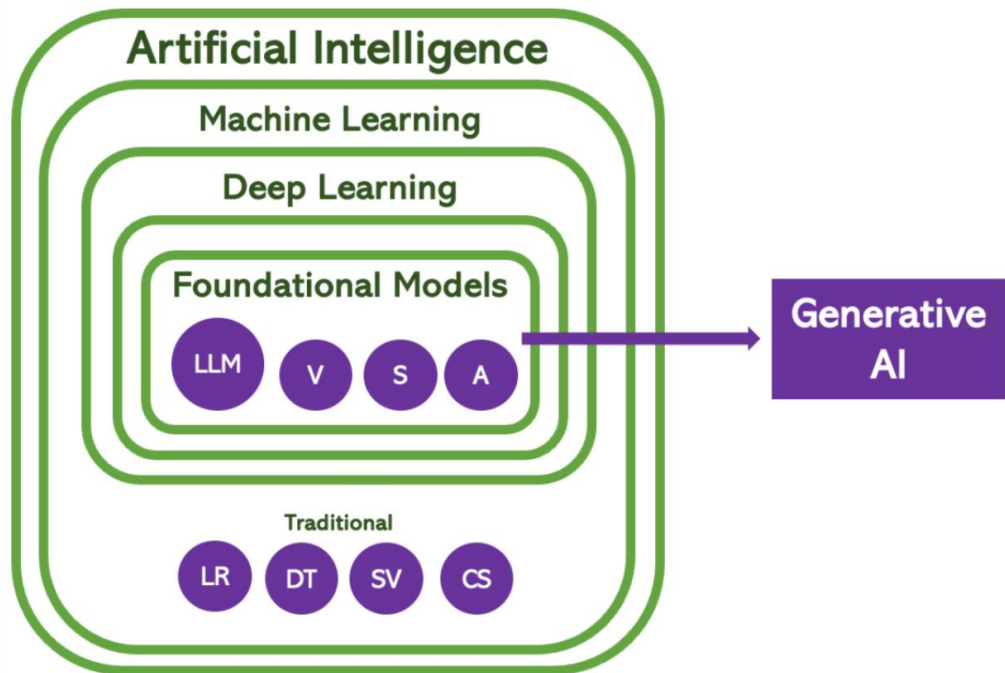
Foundational vs Traditional ML models

Foundational	Traditional ML
• Trained on broad unlabeled data for a variety of tasks • Very large number of parameters • Fine-tuning for adaptation • Self-supervision for labeling • Reduced development time, less ongoing maintenance • Dataset bias • lack of domain specificity, potential for misuse	• Trained on specific datasets and designed for particular tasks • Fewer parameters compared to foundation models. • Label supervision • Easy to implement, interpret, and computationally efficient. • Risk of overfitting or underfitting • Limited generalization

How has the landscape changed?

Traditional ML	Foundational models
Engineered features	Auto-derived features
Novel architectures	Mostly transformer variants.
Multiple independent instances of the product	Single instance adapted
Trained with smaller, hand-selected and supervised datasets	Trained with self-supervised on very large datasets
Trained for a specific purpose, often classification.	Trained once and applied to different use cases.
Less compute resources	Power hungry GPU-dependent
Deployment was simple as library calls	Full ML Ops behind training, fine-tuning and deployment.
Humans focused on design and inventions of architectures, less focus on data	More focus on data since architectures are standardized. Most improvements are practical with less radical changes.
Less focus on data collection	A whole industry of annotators, alignment, and LLM training data collection

Evolution of ML models

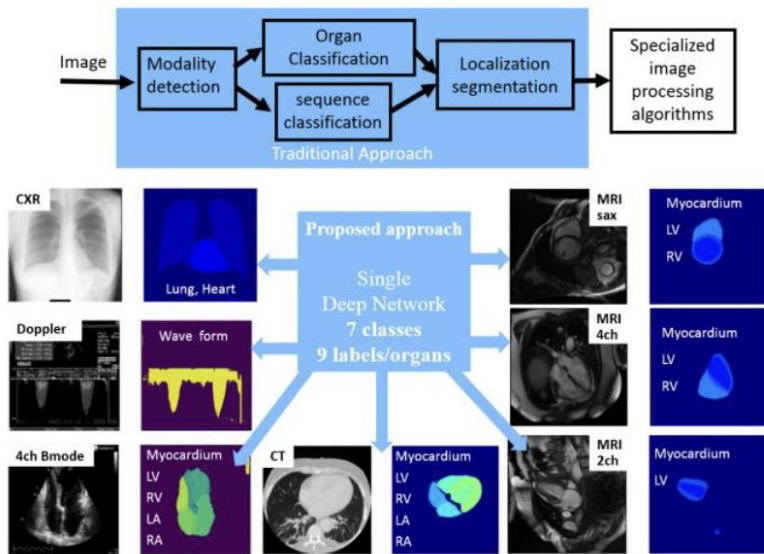


<https://www.linkedin.com/pulse/ai-alphabet-soup-from-buzzwords-brilliance-brian-gruttadauria-fxtie/>

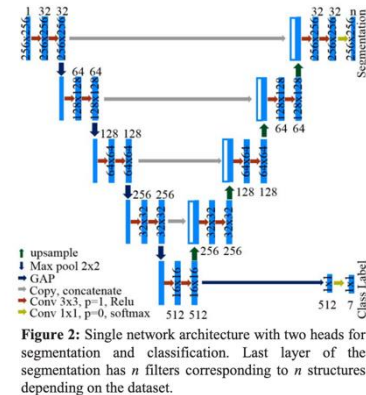
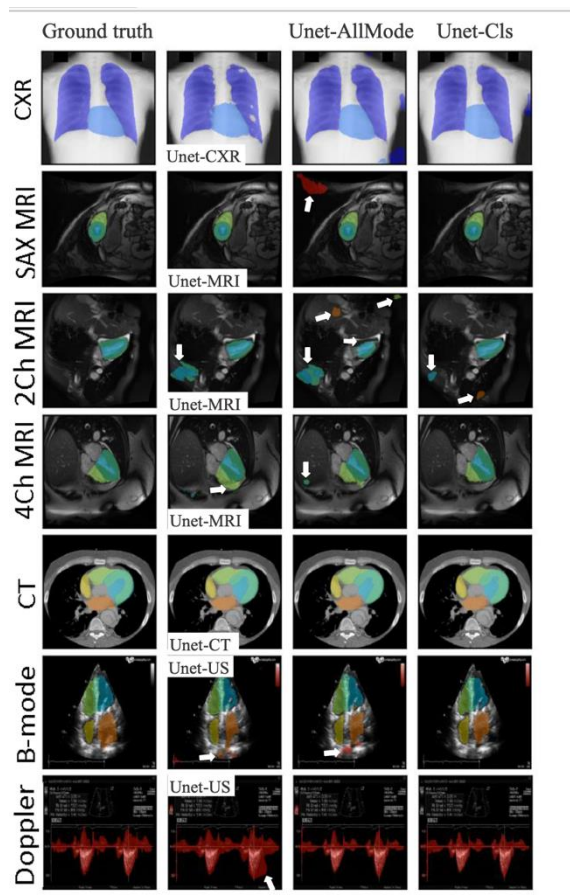
Foundation Models in Medical Imaging

- What "foundation" means for medical imaging
 - breadth of data/tasks/modalities, not simply parameter count.
 - Pretraining vs. downstream adaptation
 - Foundation model vs. large pretrained model vs. generic backbone
- Why medical imaging is unusual:
 - enormous 3D volumes
 - multiple modalities
 - limited labeled data
 - institution/scanner/population shift
 - highly heterogeneous anatomy and pathology

Early DL segmentation models were foundational in medical imaging



As early as ISBI 2018!



Foundation models to the rescue!



Key ideas in foundational models for medical imaging

- Transformers
 - Tokenization
 - Attention
 - Encoders
 - Decoders
- Representation learning
 - Supervised (fully-labeled)
 - Semi-supervised (labeled + unlabeled)
 - Self-supervised (auto-labeled data, using input as output)
- Vision-only models
- Vision-language models
- Segmentation models
- Image generation models
- Main tasks
 - Classification
 - Segmentation
 - Open vocabulary learning
 - Report generation
 - Report verification

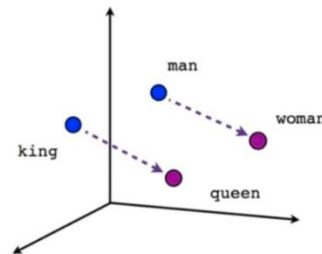
Masked language prediction – A key element of foundational modeling

I would like to say a few more words about the
ADJECTIVE
most important invention of the twentieth century. I am not
referring to science or even to the discovery of
AN INVENTION
science. The most important invention,
A FOOD ADJECTIVE
in my opinion, is the sneaker. If it were not for sneakers, our
lives would be dirty, cold, and
PART OF BODY (PLURAL)
miserable. Sneakers keep me from skidding if the
ADJECTIVE
rocks are slippery, and when I run, they keep me
PLURAL NOUN
from stubbing my foot.
PLURAL NOUN

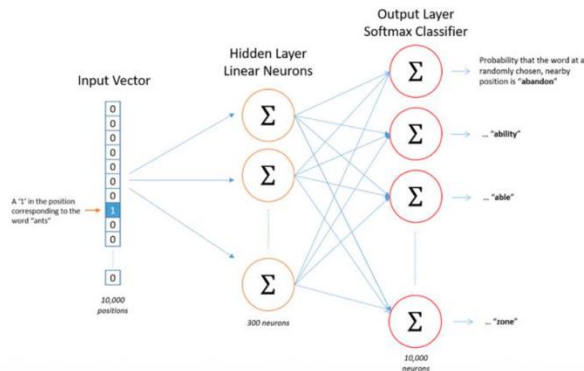
(source: Medium article 2018)

Word2Vec model

- From Words to numeric vectors
- 2-layer NN
- Input : one-hot vector of vocabulary
- Hidden layer : A vector per vocabulary word serving as the embedding
- Output: predicted probabilities of seeing another word within distance k from the input word
- Captures both syntactic and semantic similarity



Demo



Understanding the neural network training of Word2Vec model

Transformers

- Your kitchen sink of good ideas all rolled into one architecture
 - Tokenizing the input text
 - Positional encoding of tokens to preserve order
 - Attention – the Q,K,V paradigm
 - Multi-head attention – for more combinations to explore
 - Multi-layer encoders for increased abstractions
 - Linear layers for aggregating it all together
 - Explores this in parallel for all tokens unlike RNN
 - Representations created more powerful than CNNs!

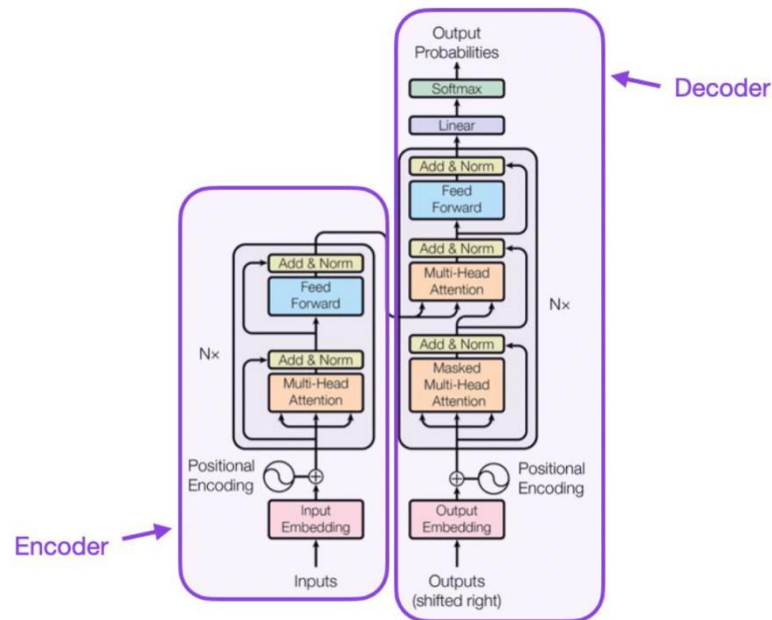


Figure 1: The Transformer - model architecture.

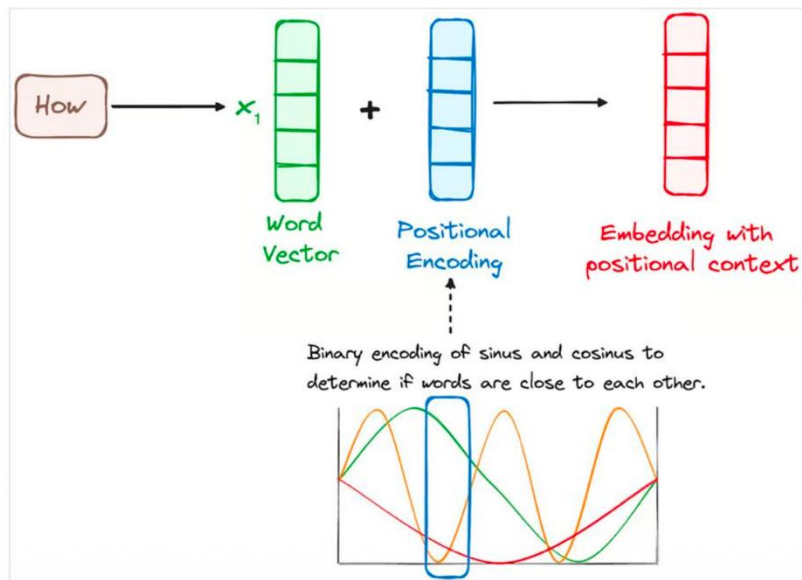
<https://www.linkedin.com/pulse/lm-transformer-architecture-shivasish-mahapatra-kj9qf/>

Tokenizing the input

- Word2Vec encoding of base vocabulary
 - 30,000+ words
- Word-piece tokenization for handling out-of-vocabulary
- E.g. A sequence of numbers or letters that opens a combination lock
 - ['<s>', 'a', 'Ġsequence', 'Ġof', 'Ġnumbers', 'Ġor', 'Ġletters', 'Ġthat', 'Ġopens', 'Ġa', 'Ġcombination', 'Ġlock', '</s>']
- This combination of terms makes sense to me to describe this beautifying phenomenon .
 - ['<s>', 'This', 'Ġcombination', 'Ġof', 'Ġterms', 'Ġmakes', 'Ġsense', 'Ġto', 'Ġme', 'Ġto', 'Ġdescribe', 'Ġthis', 'Ġbeaut', 'Ġifying', 'Ġphenomenon', 'Ġ.', '</s>']
- Tokens may be non-sensical from language perspective

Capturing sequence information

- Allows for sequence and nearness of words.
- But is additive operation the only possibility?



$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

Attention

- What is (Q,K,V)
 - From Information retrieval literature
 - Key is the index or pointer to the content, q =query, V =actual value stored
 - Also a model for human memory retrieval
 - In the context of sequence modeling:
 - Capturing relations between words.
 - Given a word Q , and its neighbors x -steps away K , can you predict the likelihood of a word V ?
 - In Word2Vec we did a fixed K (e.g. immediate neighbor)
 - More general masked language modeling

Attentional computing

- Predominantly based on modeling context by exploring positional relationships between words in sentences

e.g. Hello, how are you ?

Vocabulary

$V = \{\text{"Hello", ",","how", "are", "you", "?"}\}$

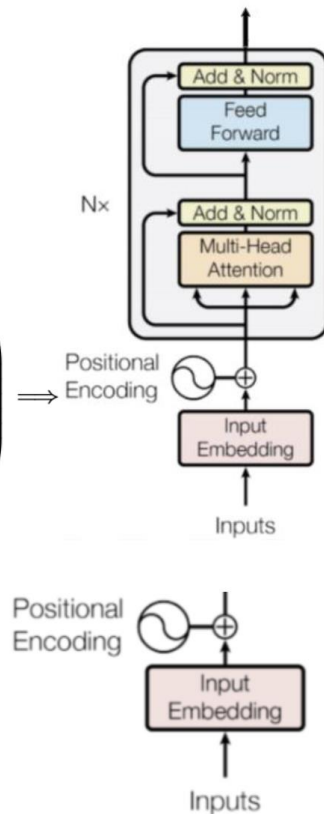


More complex relationships and context modeling

Hello, how are you ?

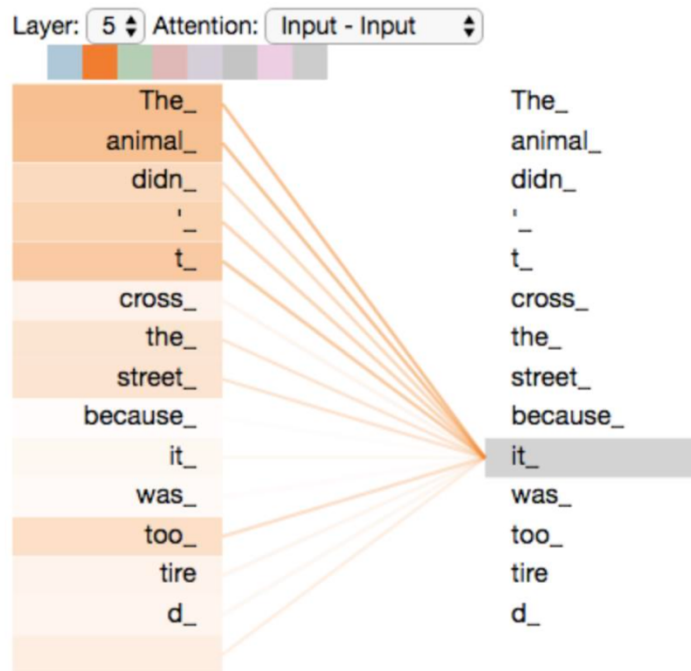
	<i>Hello</i>	,	<i>how</i>	<i>are</i>	<i>you</i>	?
<i>Hello</i>	78.49	43.29	1.2	41.74	91.43	74.47
,	95.84	28.78	57.13	68.20	-60.94	26.85
<i>how</i>	-95.69	-52.16	17.00	45.71	48.49	64.35
<i>are</i>	-69.92	85.16	94.94	91.04	-92.83	77.49
<i>you</i>	65.85	55.85	62.54	-97.46	76.38	13.20
?	-30.05	-4.52	76.02	42.35	15.29	63.61

	<	-	d_{emb_dim}	-	>
<i>Hello</i>	123.4	0.32	...	94	32
,	83	34	...	77	19
<i>how</i>	0.2	50	...	33	30
<i>are</i>	289	432.98	...	150	92
<i>you</i>	80	46	...	23	32
?	41	21	...	74	33



Attention

e.g. I am going to my home and play with toy house.



The output embedding for “house” is mainly attending to “toy” and “play”, while the output embedding for “home” is attending to “my” and “going”. Even though home and house are synonyms and more related.

$$\text{Attention}(q, k, v) = \overbrace{\text{softmax}\left(\frac{qk^T}{\sqrt{d_k}}\right)}^{\text{Attention weights}} v$$

from to

vector dimensionality of K, V

<https://jalammar.github.io/illustrated-transformer/>

How is this different from cosine similarity?

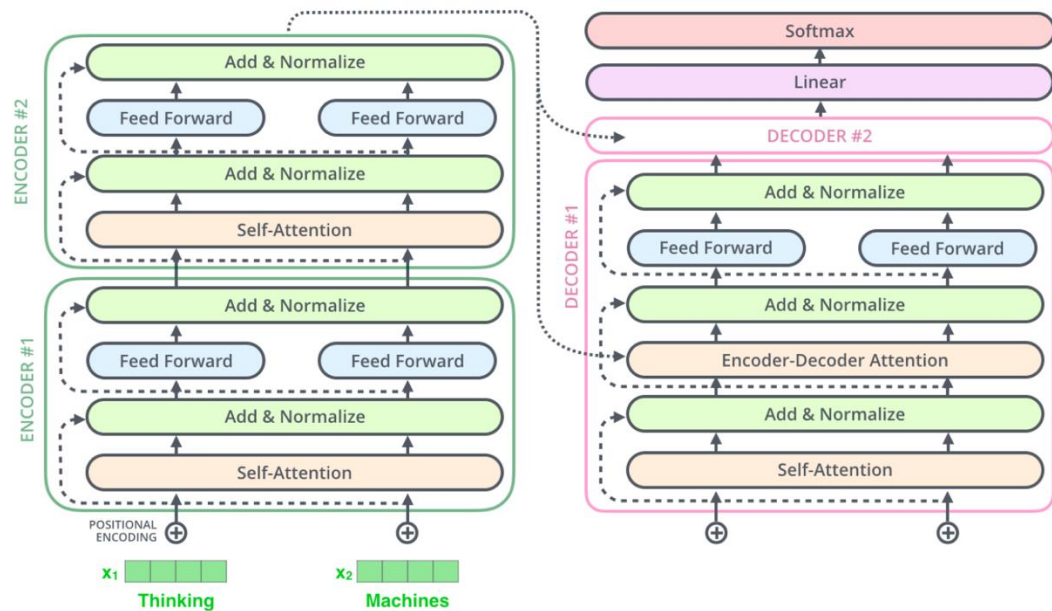
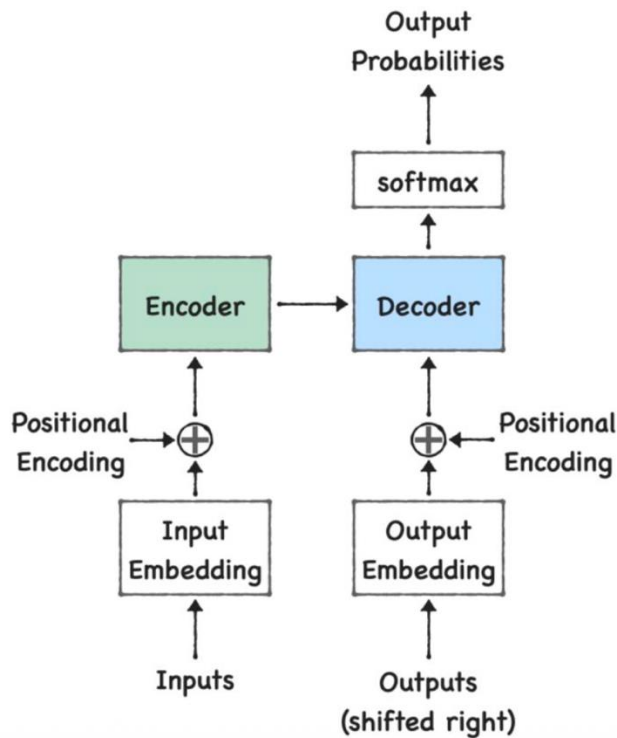
Multi-layer encoders

- Why multi-layer when we are already doing multi-head attention?
 - With multiple attention heads, each head learns a separate aspect of the data
 - Multiple layers build progressively more complex and abstract representations of the input sequence. Each layer in the encoder builds upon the previous one, refining the representations and capturing richer contextual information.
 - Similar to CNNs going over larger and larger areas of the image
 - By stacking multiple layers, the model can capture context not just at the local level, but also at a more global level, understanding how different parts of the sequence relate to each other
 - The first layer gives context for the actual adjacent.
 - And the feedforward layer after that attention layer recognizes phrases.
 - The next attention layer gives context to phrases about other phrases, so that “threw the ball” and “over the wall” get paired together.
 - And then higher and higher features get recognized each layer and the next attention contextualizes those

Why linear layers or feed-forward layers?

- Map input representations to output predictions
- Transforming data within the model's architecture.
- Project vectors into different dimensions
- Enables the model to learn complex relationships, transform data, and generate meaningful output

Summarizing the transformer model



Can use encoder only - BERT
Decoder only - GPT-3
Encoder-decoder models - GPT-4,
Llama, ec.

Figure 1: The original encoder-decoder architecture of a transformer model (adapted from Vaswani et al, 2017)

Vision Transformer

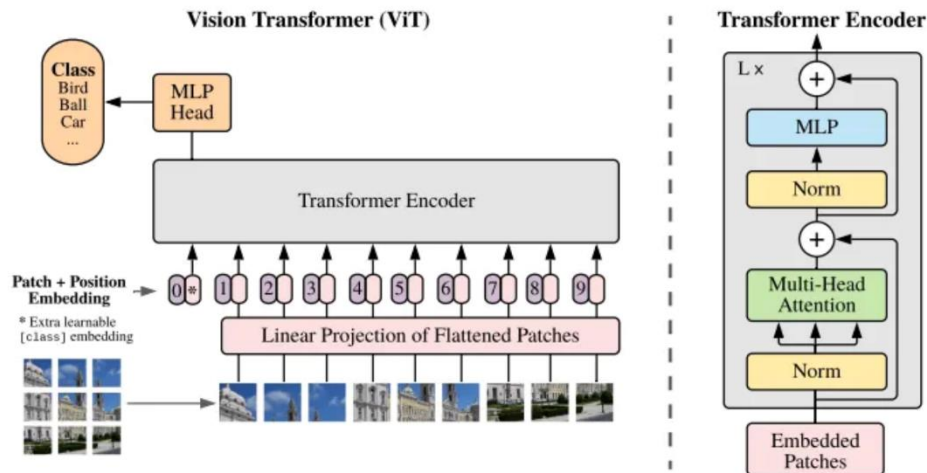


Figure 1: Vision Transformer Model Overview. Image: [ViT Paper](#).

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Table 1: Details of Vision Transformer model variants.

Additional considerations:

- What is a token?
- How to encode the token?
- 2D positional embeddings
- Number of layers
-

Published as a conference paper at ICLR 2021

AN IMAGE IS WORTH 16X16 WORDS:
TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy^{*,†}, Lucas Beyer^{*}, Alexander Kolesnikov^{*}, Dirk Weissenborn^{*}, Xiaohua Zhai^{*}, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*,†}

^{*}equal technical contribution, [†]equal advising

Google Research, Brain Team

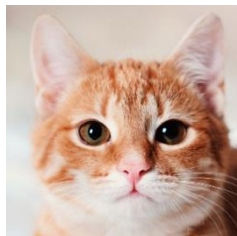
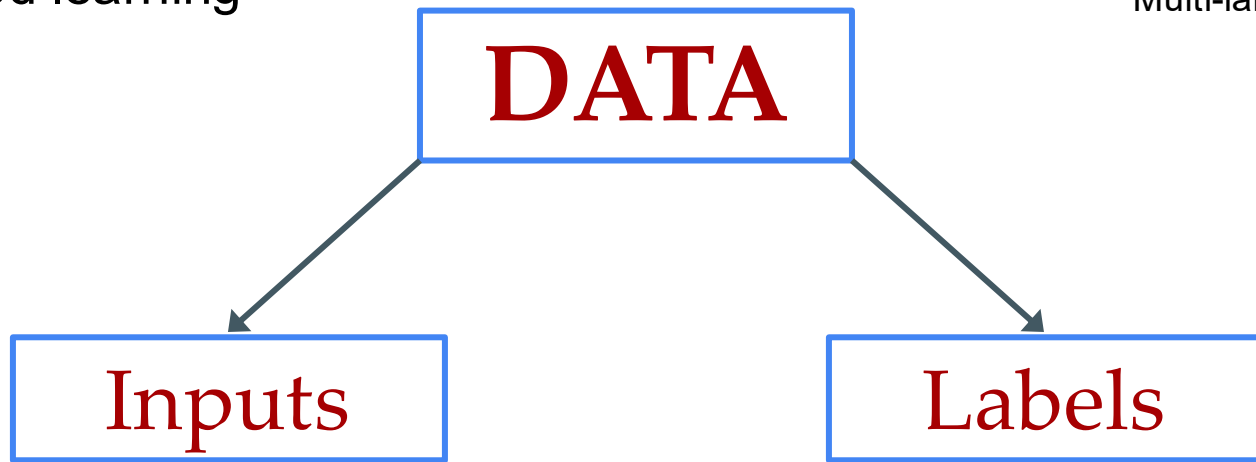
{adosovitskiy, neilhoulshby}@google.com

Representation learning

- Supervised learning
 - Original VIT was a supervised classifier for objects within
- Semi-supervised learning
- Curriculum learning
- Transfer learning
 - Knowledge distillation
- Self-supervised learning

Supervised learning

Multi-class classification
Multi-label classification



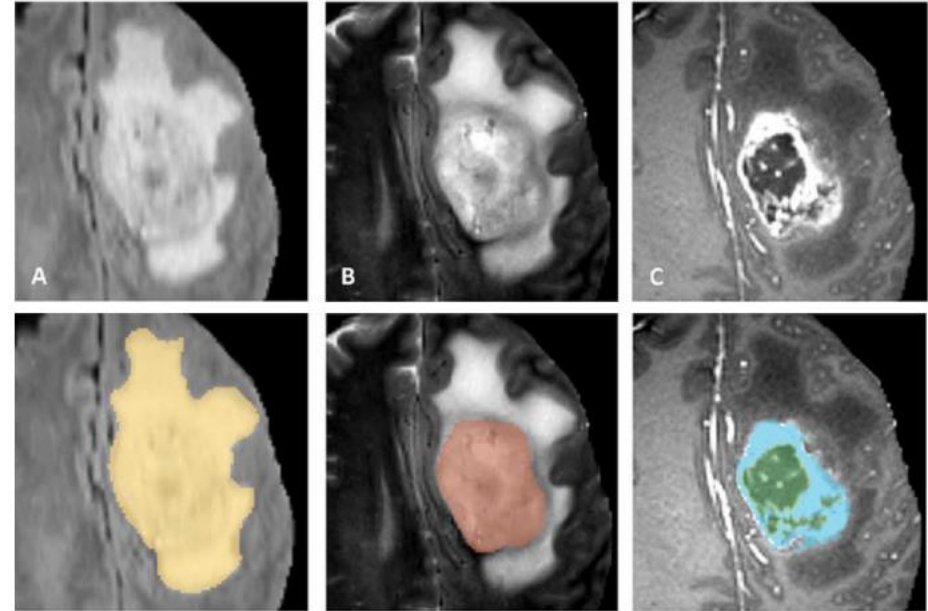
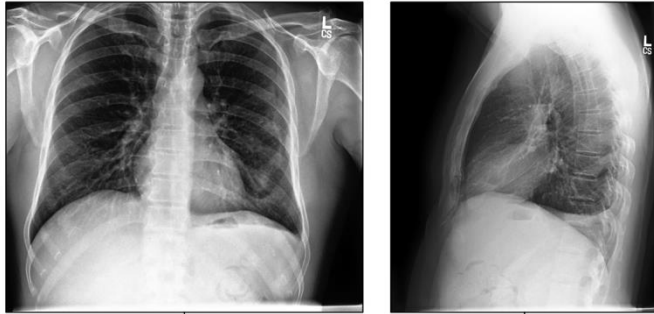
Dog
Cat
Hot Dog

and so on...

...

Create representations -> Feature vector -> Classifier

Example Labels in Medical Imaging



[1] Irwin et al. CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison. AAAI. 2019

[2] Menze et al. The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). IEEE TMI. 2015

Semi-supervised learning

- Learning from partial data
 - Learn from small -> predict for unlabeled -> retrain using pseudo labels -> Test improved representation on original small data
 - Select from psneudo labels
 - High confidence ones
 - Smoothness constraint
 - correct pseudo-labels outnumber the incorrect ones
 - Augmentation stress test
 - Easy augmentations (flips) to hard augmentations (crops) but since the identity is the same, learn a robust prior
 - Regularization

Beyond Supervised Learning

- Semi-Supervised Learning

Step 1:

IM1	IM2
IM4	IM5

+

L1	L2
L4	L5

= Model 1

Step 2:

		IM3
		IM6
IM7	IM8	IM9

Model 1

+

		PL3
		PL6
PL7	PL8	PL9

= Dataset 2

Step 3:

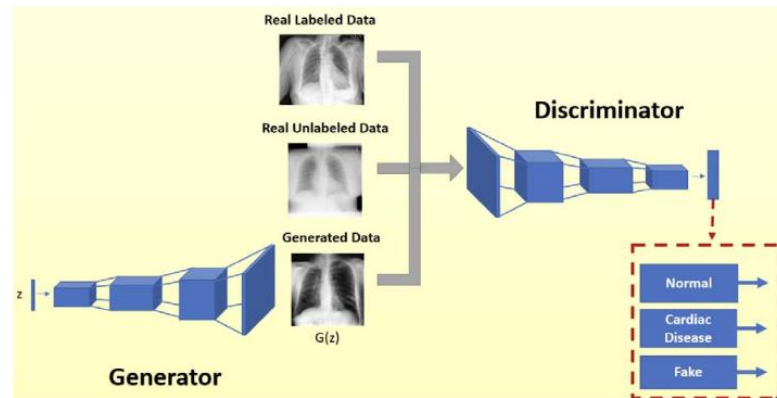
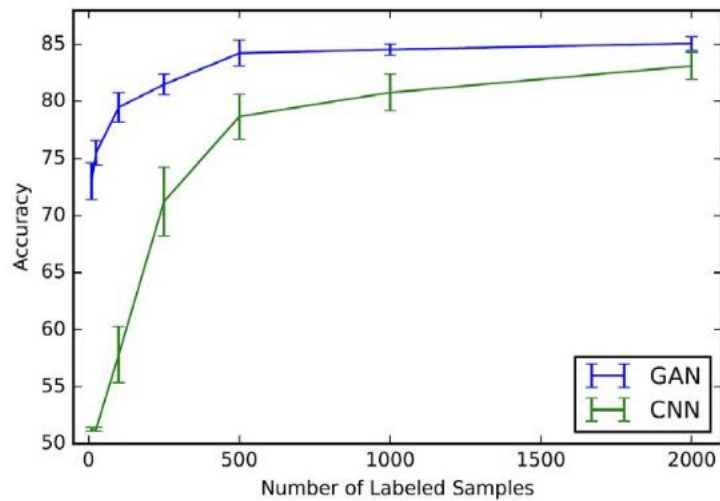
IM1	IM2	IM3
IM4	IM5	IM6
IM7	IM8	IM9

+

L1	L2	PL3
L4	L5	PL6
PL7	PL8	PL9

= Model 2

Semi-supervised learning



$$J^{(D)} = -\frac{1}{2} \mathbf{E}_{x \sim p_{data}} \log D(x) - \frac{1}{2} \mathbf{E}_z \log(1 - D(G(x)))$$

$$J^{(G)} = -\frac{1}{2} \mathbf{E}_z \log D(G(x))$$

$$L = L_{labeled} + L_{unlabeled} + L_{generated}$$

$$L_{labeled} = -\mathbf{E}_{x, y \sim p_{data}} \log p_{model}(y|x, y < K + 1)$$

$$L_{unlabeled} = -\mathbf{E}_{x \sim p_{data}} \log(1 - p_{model}(y = K + 1|x))$$

$$L_{generated} = -\mathbf{E}_{x \sim G} \log p_{model}(y = K + 1|x)$$

Curriculum learning

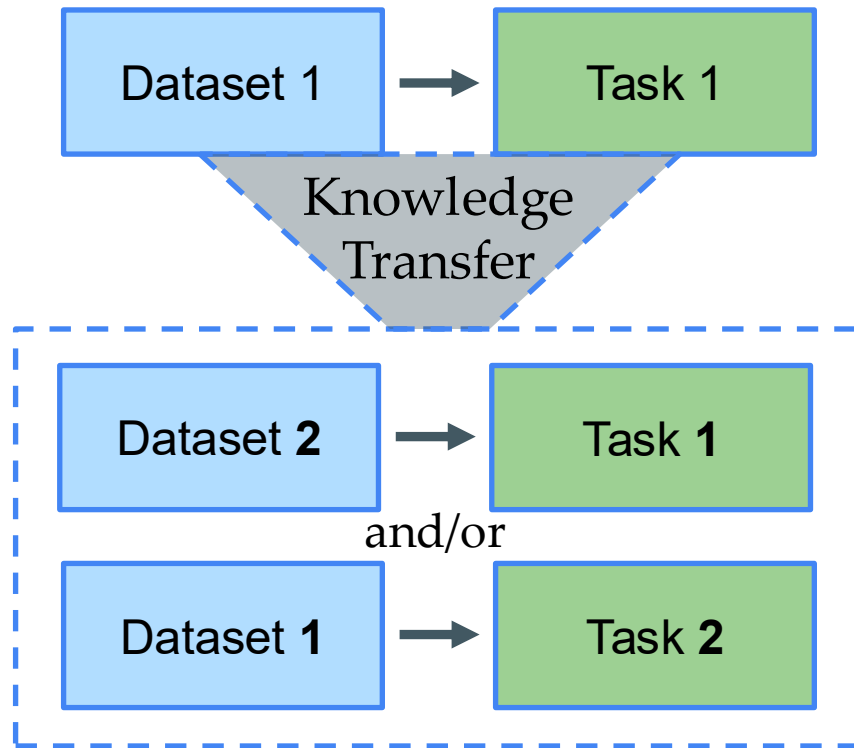
- Curriculum learning
 - Learn a simple task and move to more complex
 - organize the data and rank from easy to hard
 - Set a schedule to introduce hardness
 - Fast convergence

Transfer learning

- Transfer learning
 - Start with a pre-trained base ->freeze the lower layers (feature extractor) ->replace classification head ->fine-tune e-to-e



Transfer Learning

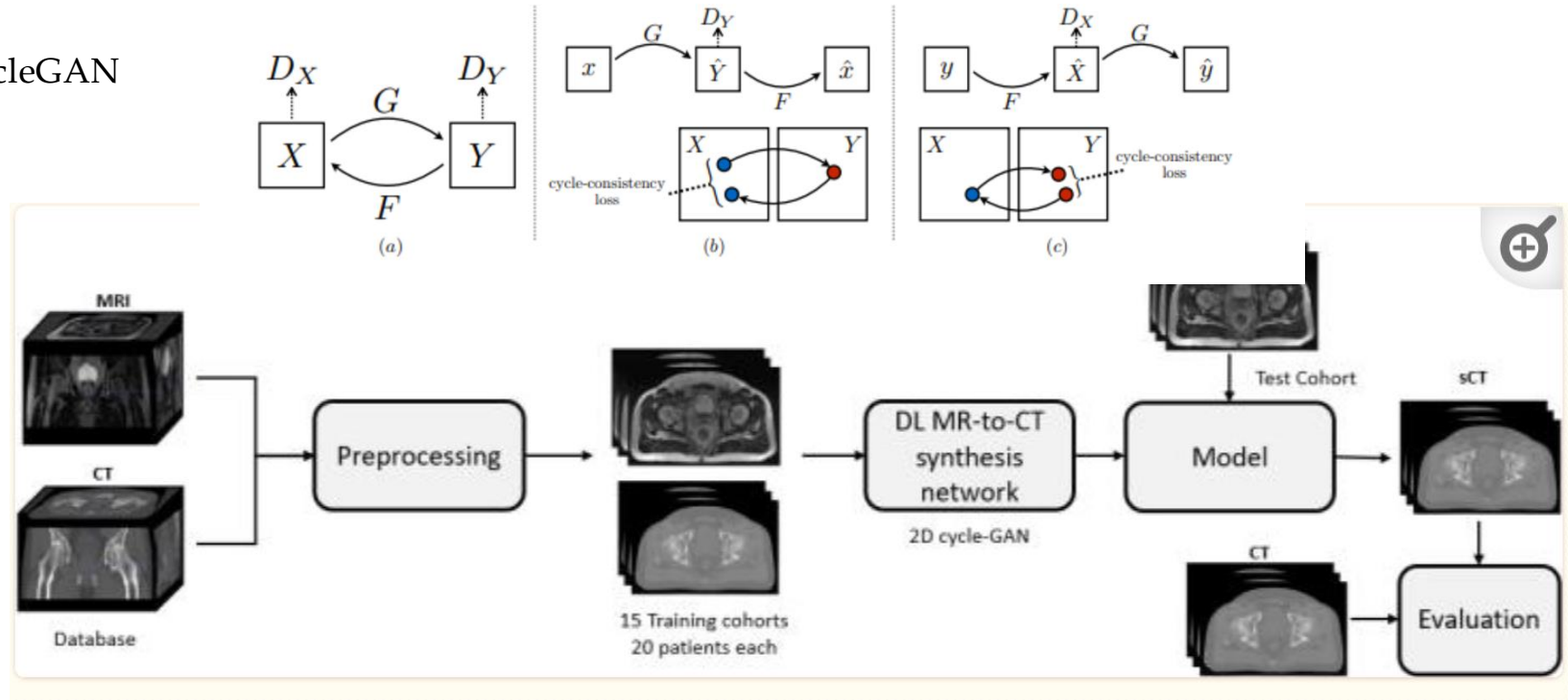


Self-Training Terminology

- **Teacher Network:** Network that creates pseudo labels
- **Student Network:** Network that learns using pseudo labels
- **Self-Training:** When student network is same/larger sized than teacher network
 - Larger number of parameters, more data augmentation, random noise
- **Knowledge Distillation:** When student network is smaller than teacher network
 - In a GAN formulation, the generator learns from the discriminator which is a larger network
- **Knowledge Adaptation:** When networks are similarly trained in multiple domains
 - Leveraging knowledge from each other
 - Learn where they are not performing as well
- **Knowledge Enhancement:**
 - Continue to train student networks with more generalization

Transfer learning in medical imaging

CycleGAN



Computed tomography synthesis from magnetic resonance imaging using cycle Generative Adversarial Networks with multicenter learning

Self-supervised learning

- Auto-encoders
 - Variational auto-encoders
- Pre-text task methods
 - In-painting
 - Remove portion -> learn to fill-in from neighbors -> learns the underlying representation
 - Guess patch, jigsaw, rotations
- Contrastive learning
- Self-distillation (e.g DINO)
 - passes different global and local crops of an image to a student and teacher network.
 - The student must predict the output of the teacher.
 - the teacher network is dynamically built directly from the student's own history.
- Masked vision modeling
 - Masked auto-encoders
- JEPA (from pixels to concepts)
 - Learning in embedding space

Traditional Reconstruction Era (Early 2010s)

Pre-text Task Era (2016 – 2018)

Contrastive Learning Boom (2019 – 2020)

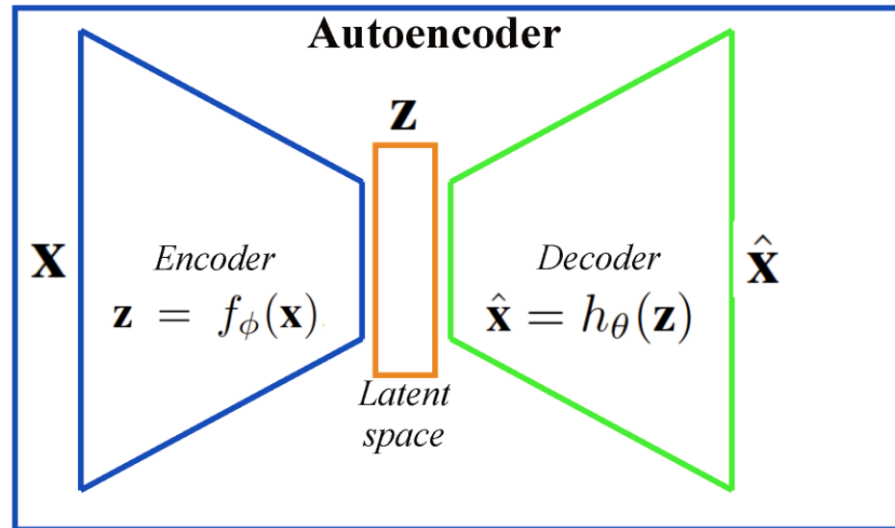
Early Self-Distillation Era (2020 – 2021)

Masked Vision Modeling Era (Late 2021 – 2022)

The Latent World Model Era (2022 – Present)

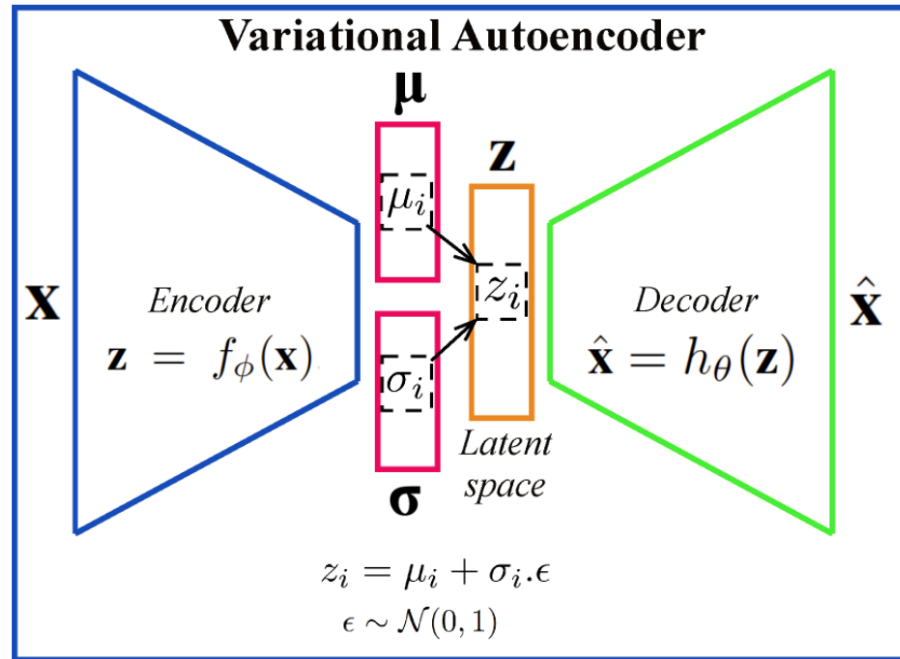
Generative Models – Autoencoders

- Encoder – Reduce the dimensionality of some input
- Decoder – Use condensed dimensionality to recover signal



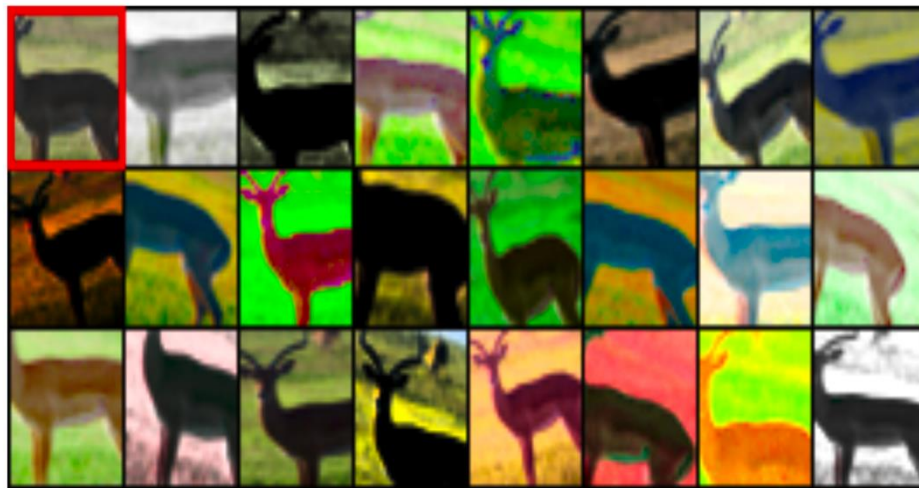
Variational Autoencoder

- Sample plausible distributions from latent space



Pretext Tasks

- The surrogate task is designed to yield generic features that are descriptive and robust to typical variations in the data.
 - The variation is simulated by randomly applying transformations to a 'seed' image.
 - So data augmentation is also a feature representation
 - Feature representation learned by the Exemplar-CNN performs well on very different tasks:
 - e.g. object classification and descriptor matching.
- Trained with unlabeled data

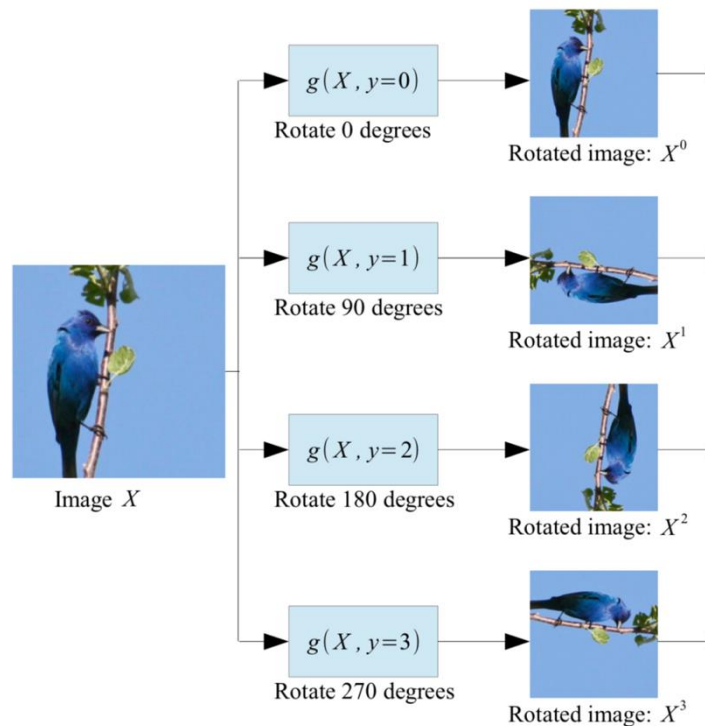


$$L(X) = \sum_{\mathbf{x}_i \in X} \sum_{T \in \mathcal{T}_i} l(i, T\mathbf{x}_i),$$

$$l(i, T\mathbf{x}_i) = M(\mathbf{e}_i, f(T\mathbf{x}_i)),$$

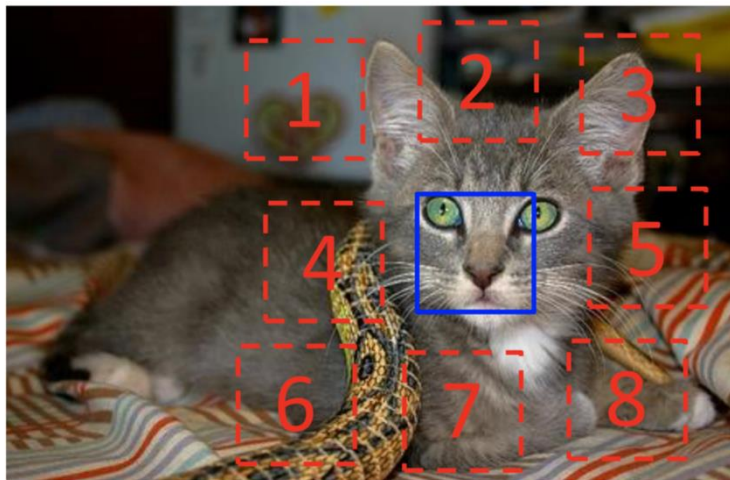
$$M(\mathbf{y}, \mathbf{f}) = -\langle \mathbf{y}, \log \mathbf{f} \rangle = -\sum_k y_k \log f_k,$$

Pretext Tasks - Rotation

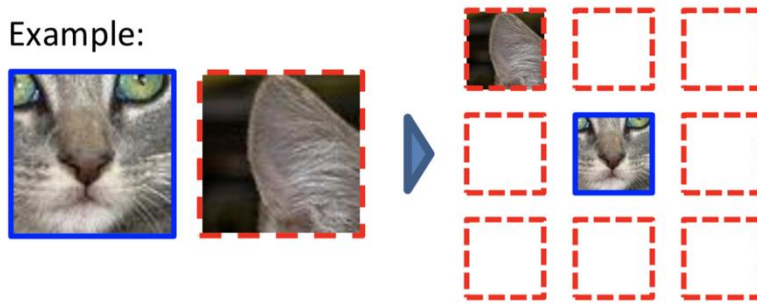


Pretext Tasks - Patching

pre- text task, i.e., predicting the relative position of patches within an image



Example:



Question 1:

Question 2:

the network must feed the two input patches through several convolution layers, and produce an output that assigns a probability to each of the eight spatial configurations (Figure 2) that might have been sampled (i.e. a softmax output).

learns a feature embedding for *individual* patches, such that patches which are visually similar (across different images) would be close in the embedding space.

What Position Does the Blue Square Occupy?

1	2	3
4	5	6
7	8	9



Image Inpainting

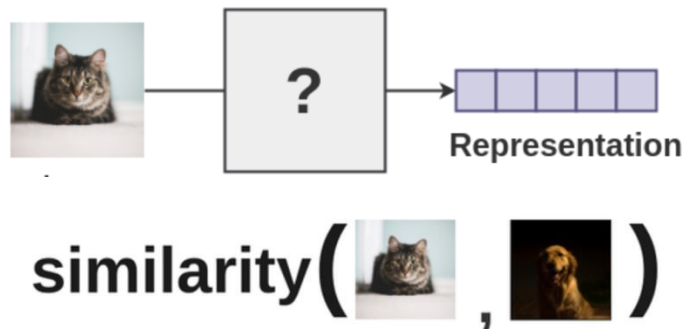
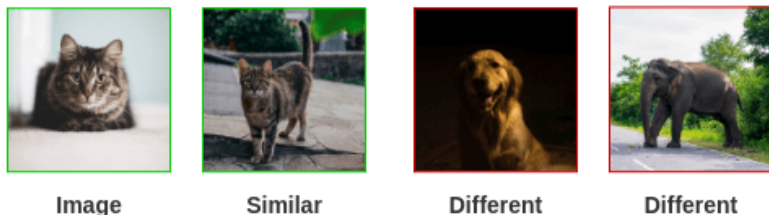
Contrastive Learning

Match the correct animal

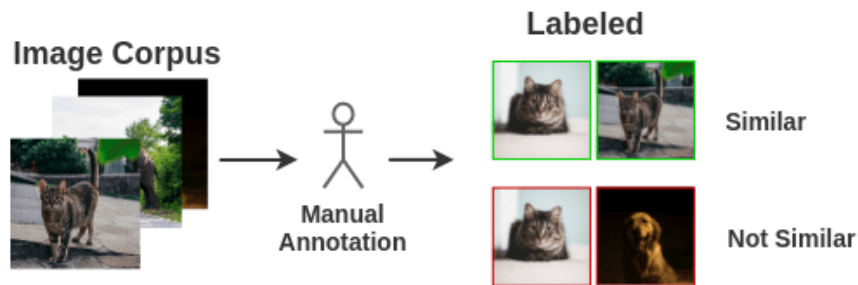


Contrastive Learning

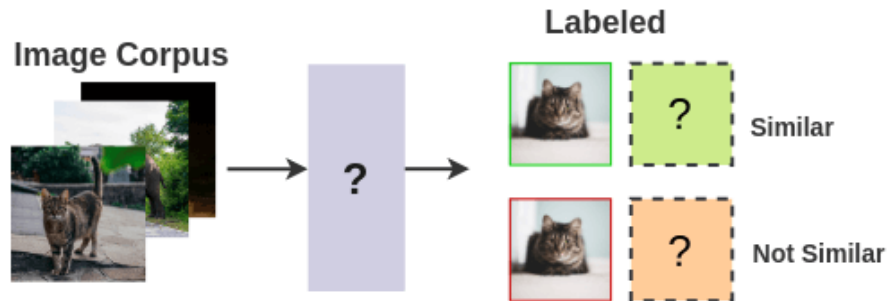
Need similar and different examples



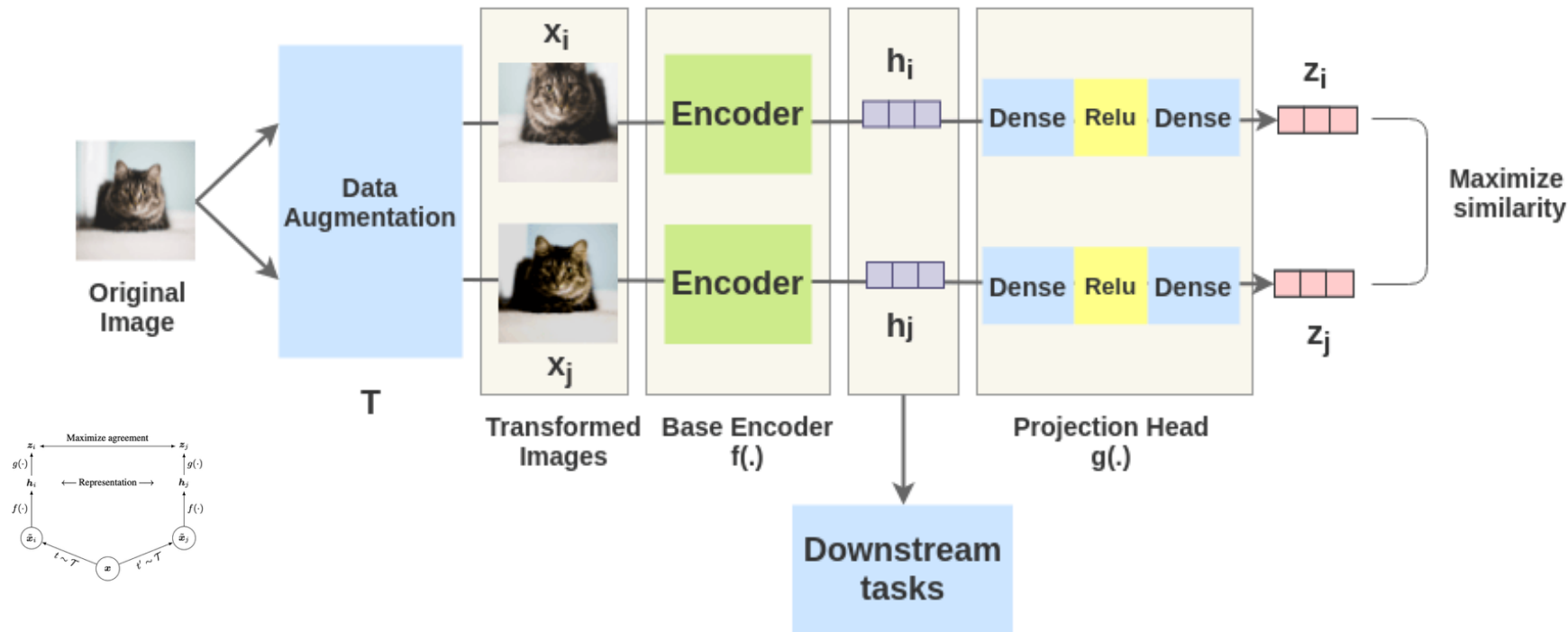
Supervised Approach



How can we automatically generate pairs?



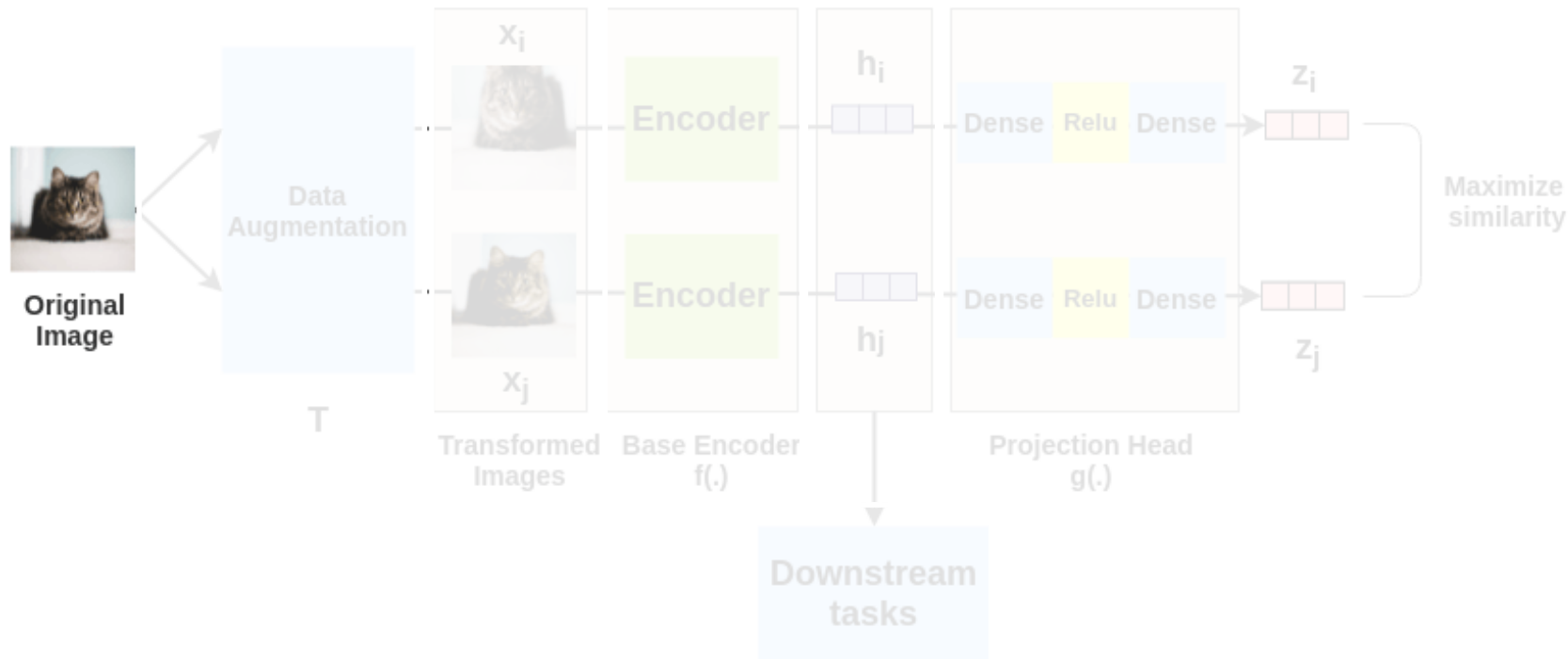
Contrastive Learning



[1] Chaudhary A. The Illustrated SimCLR Framework. <https://amitness.com/2020/03/illustrated-simclr/> 2020.

[2] Chen et al. A Simple Framework for Contrastive Learning of Visual Representations. ICML 2020.

Contrastive Learning



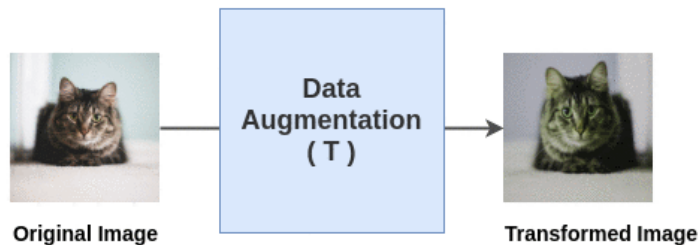
[1] Chaudhary A. The Illustrated SimCLR Framework. <https://amitnss.com/2020/03/illustrated-simclr/> 2020.

[2] Chen et al. A Simple Framework for Contrastive Learning of Visual Representations. ICML 2020.

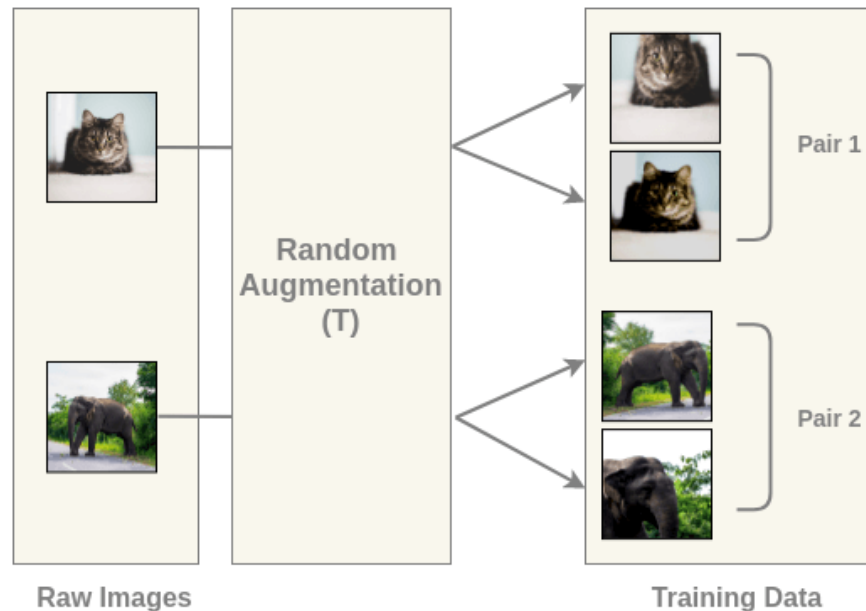
Contrastive Learning

Preparing similar pairs in a batch

Random Transformation



Batch Size
 $N = 2$



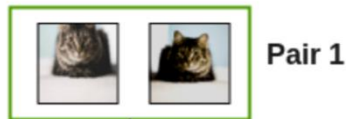
[1] Chaudhary A. The Illustrated SimCLR Framework. <https://amitness.com/2020/03/illustrated-simclr/> 2020.

[2] Chen et al. A Simple Framework for Contrastive Learning of Visual Representations. ICML 2020.

Contrastive Learning

Similarity Calculation of Augmented Images

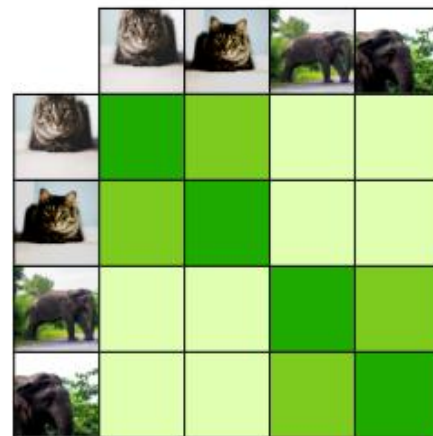
$$\text{similarity}(x_i, x_j) = \text{cosine similarity}(z_i, z_j)$$



Softmax =

$$\frac{e^{\text{similarity}(x_i, x_j)}}{e^{\text{similarity}(x_i, x_j)} + e^{\text{similarity}(x_i, x_k)} + e^{\text{similarity}(x_i, x_l)}}$$

Pairwise cosine similarity



[1] Chaudhary A. The Illustrated SimCLR Framework. <https://amitnss.com/2020/03/illustrated-simclr/> 2020.

[2] Chen et al. A Simple Framework for Contrastive Learning of Visual Representations. ICML 2020.

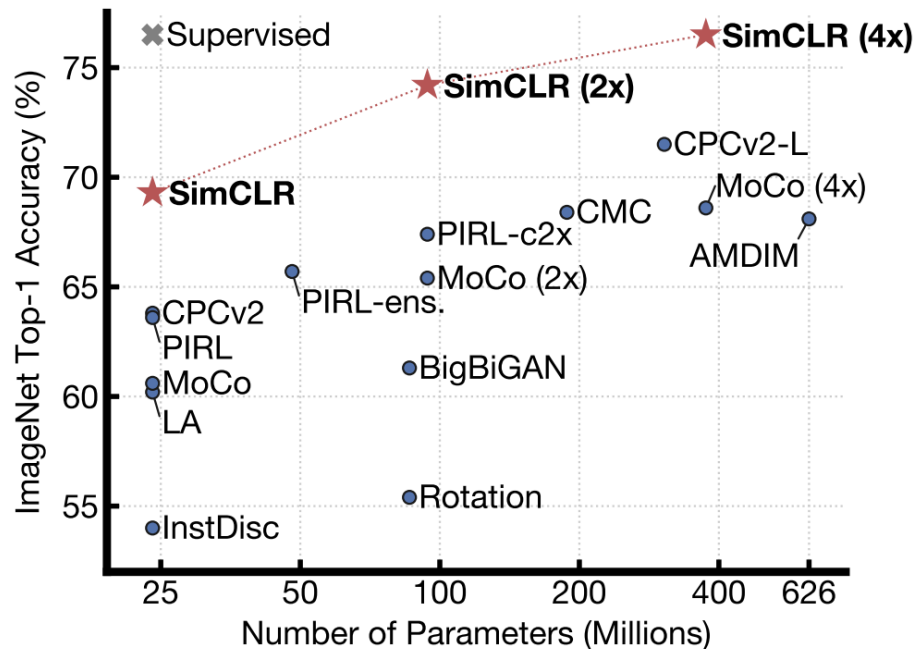
Noise Contrastive Estimation Loss

- Compute over both pairs to account for asymmetry

$$l(i, j) = -\log \frac{\exp(s_{i,j})}{\sum_{k=1}^{2N} \exp(s_{i,k})}$$

$$l(\text{cat}_1, \text{cat}_2) = -\log \left(\frac{\exp(\text{similarity}(\text{cat}_1, \text{cat}_2))}{\exp(\text{similarity}(\text{cat}_1, \text{cat}_2)) + \exp(\text{similarity}(\text{cat}_1, \text{elephant}_1)) + \exp(\text{similarity}(\text{cat}_1, \text{elephant}_2))} \right)$$

SimCLR Technique



- Large batch size requirements
- Long training times
- Heuristic data augmentations

SimCLR Augmentations



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



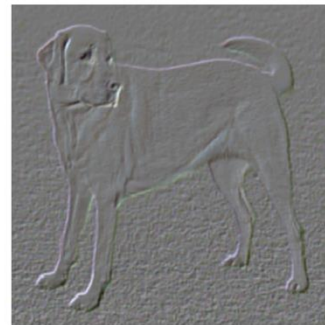
(g) Cutout



(h) Gaussian noise

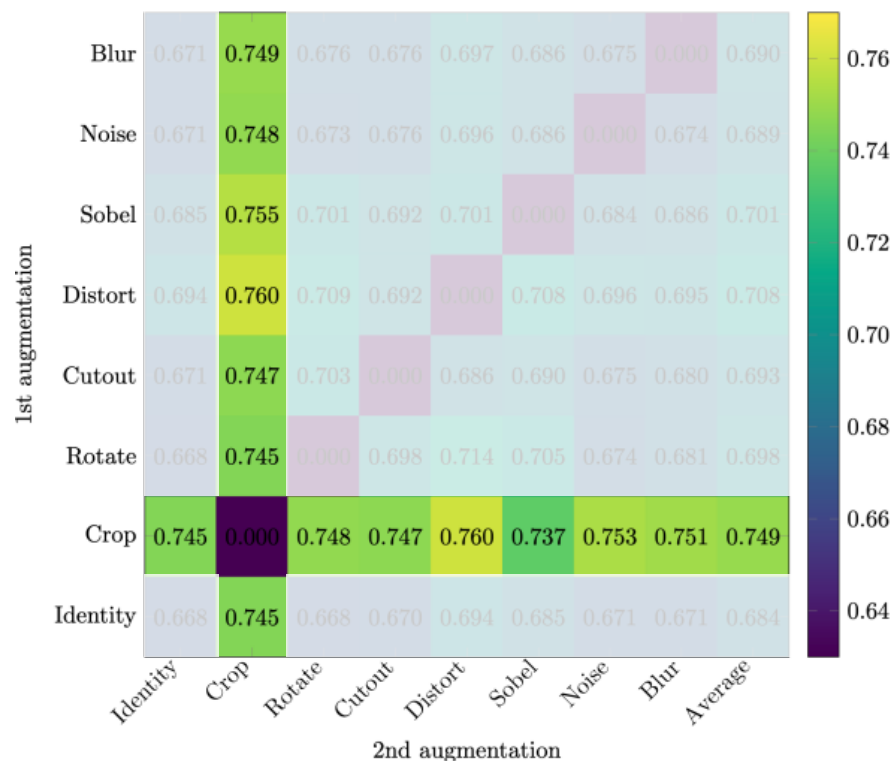
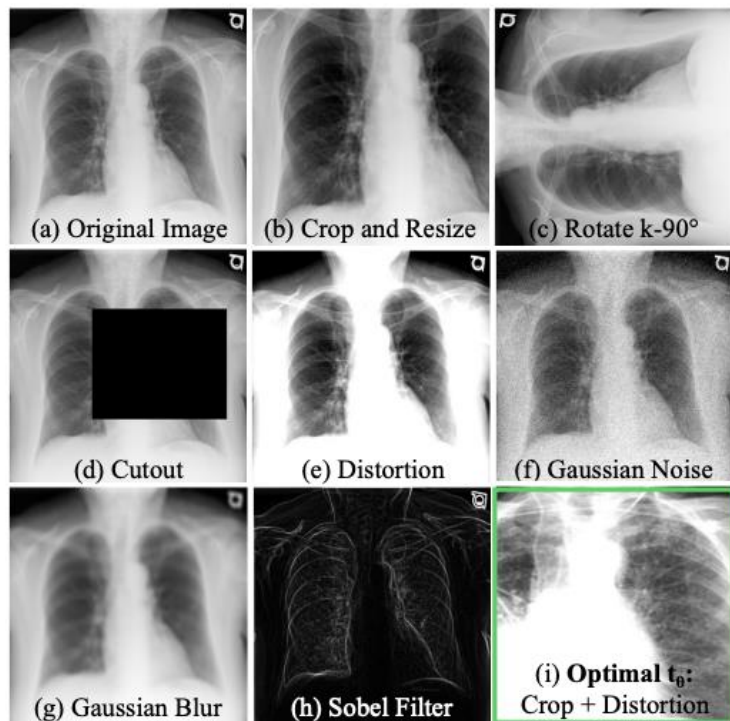


(i) Gaussian blur



(j) Sobel filtering

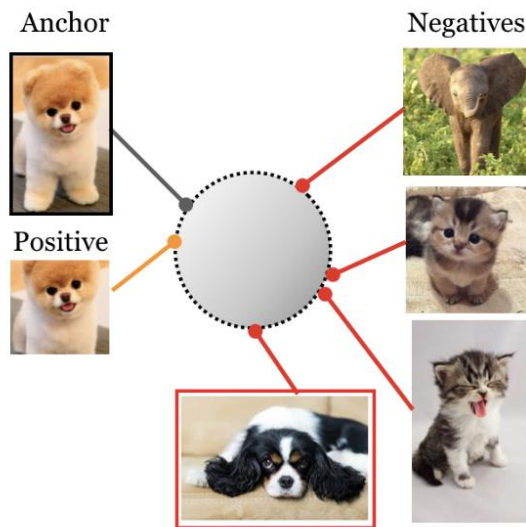
Augmentations for Medical Imaging



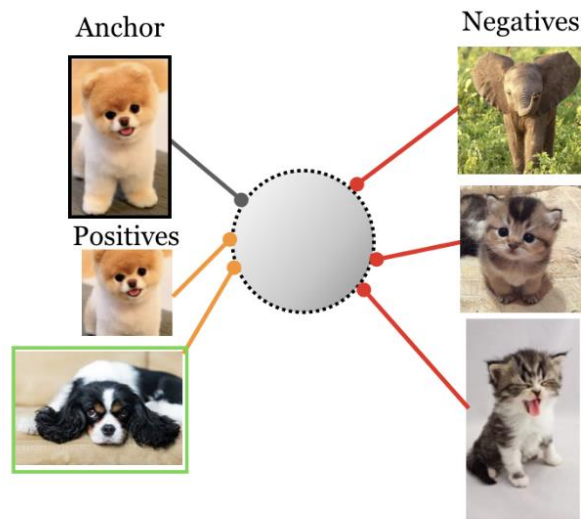
Supervised contrastive learning

- Multiple label cases
- Supervised learning

$$\mathcal{L}^{self} = \sum_{i \in I} \mathcal{L}_i^{self} = - \sum_{i \in I} \log \frac{\exp(z_i \cdot z_{j(i)} / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)}$$



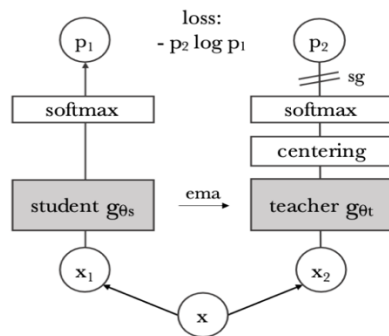
Self Supervised Contrastive



Supervised Contrastive

Self-distillation with no labels (DINO)

- The idea of **self-distillation** in DINO is to use two similar ViT networks to classify the patches.
- The **teacher** network gets the global views as an input, while the **student** network get both the local and global ones.
- Both have a MLP head to predict the softmax probabilities, but do **not** use any labels.

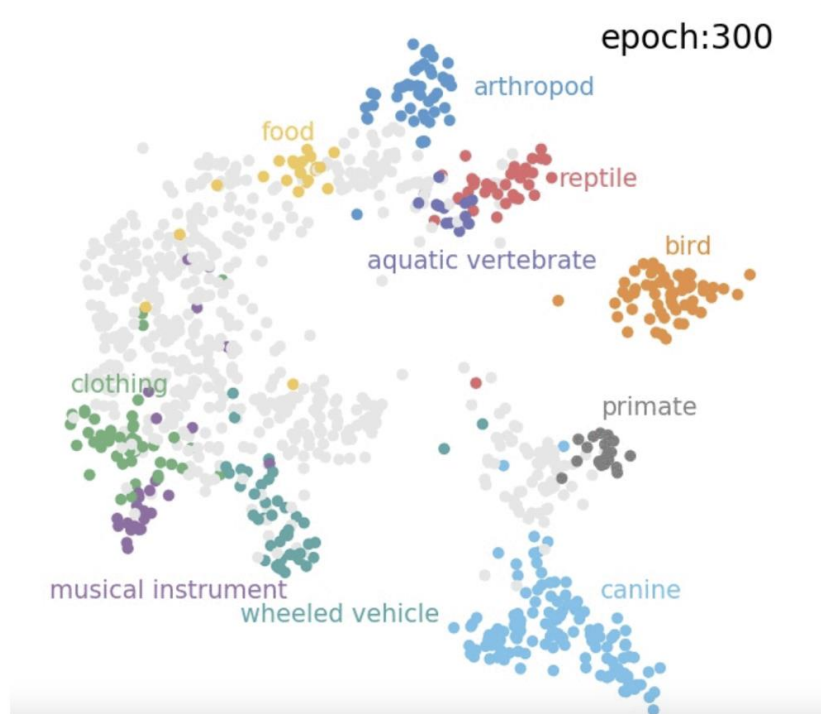


- The student tries to imitate the output of the teacher, by minimizing the **cross-entropy** (or KL divergence) between the two probability distributions.
- The teacher slowly integrates the weights of the student (momentum or exponentially moving average ema):

$$\theta_{\text{teacher}} \leftarrow \beta \theta_{\text{teacher}} + (1 - \beta) \theta_{\text{student}}$$

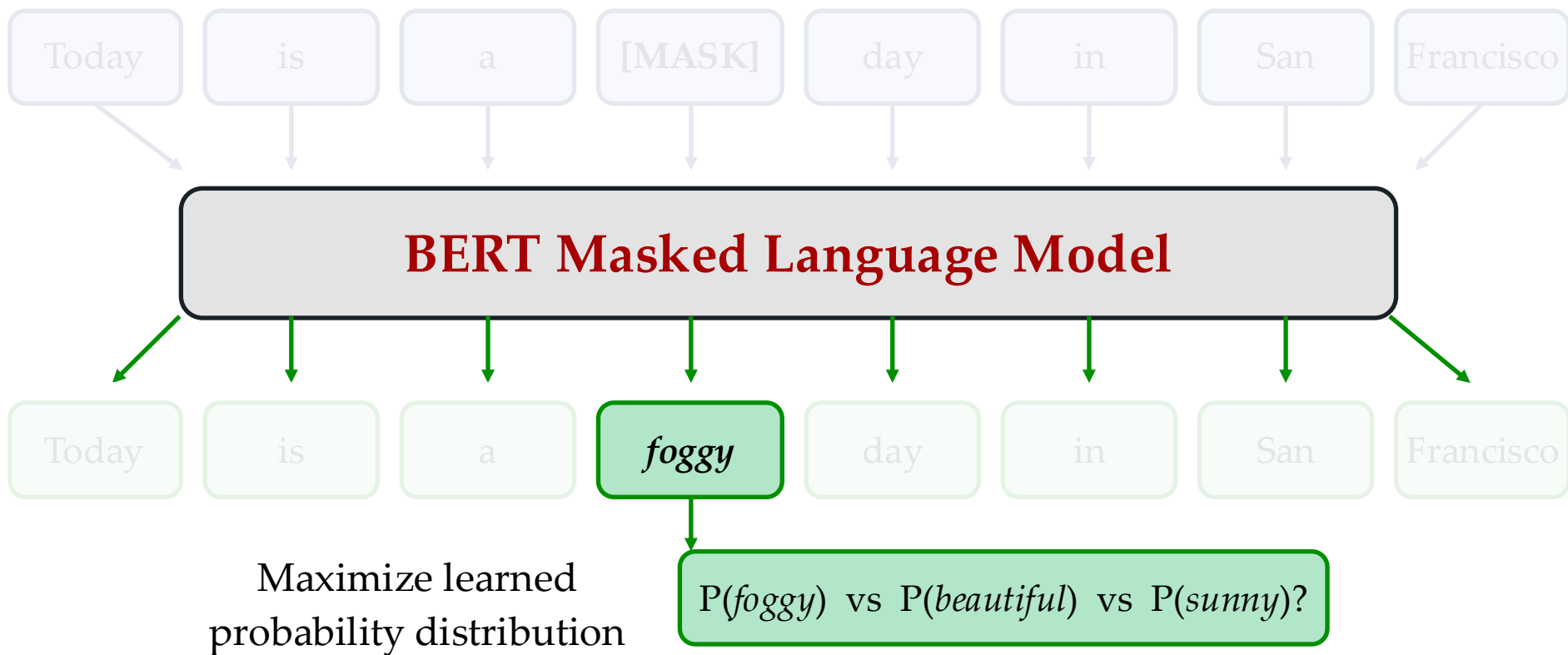
Self-distillation with no labels (DINO)

- The predicted classes do not matter when pre-training, as there is no ground truth.
- The only thing that matters is the **high-level representation** of an image before the softmax output, which can be used for transfer learning.
- Self-distillation forces the representations to be meaningful at both the global and local scales, as the teacher gets global views.
- ImageNet classes are already separated in the high-level representations: a simple kNN (k-nearest neighbour) classifier achieves 74.5% accuracy (vs. 79.3% for a supervised ResNet50).



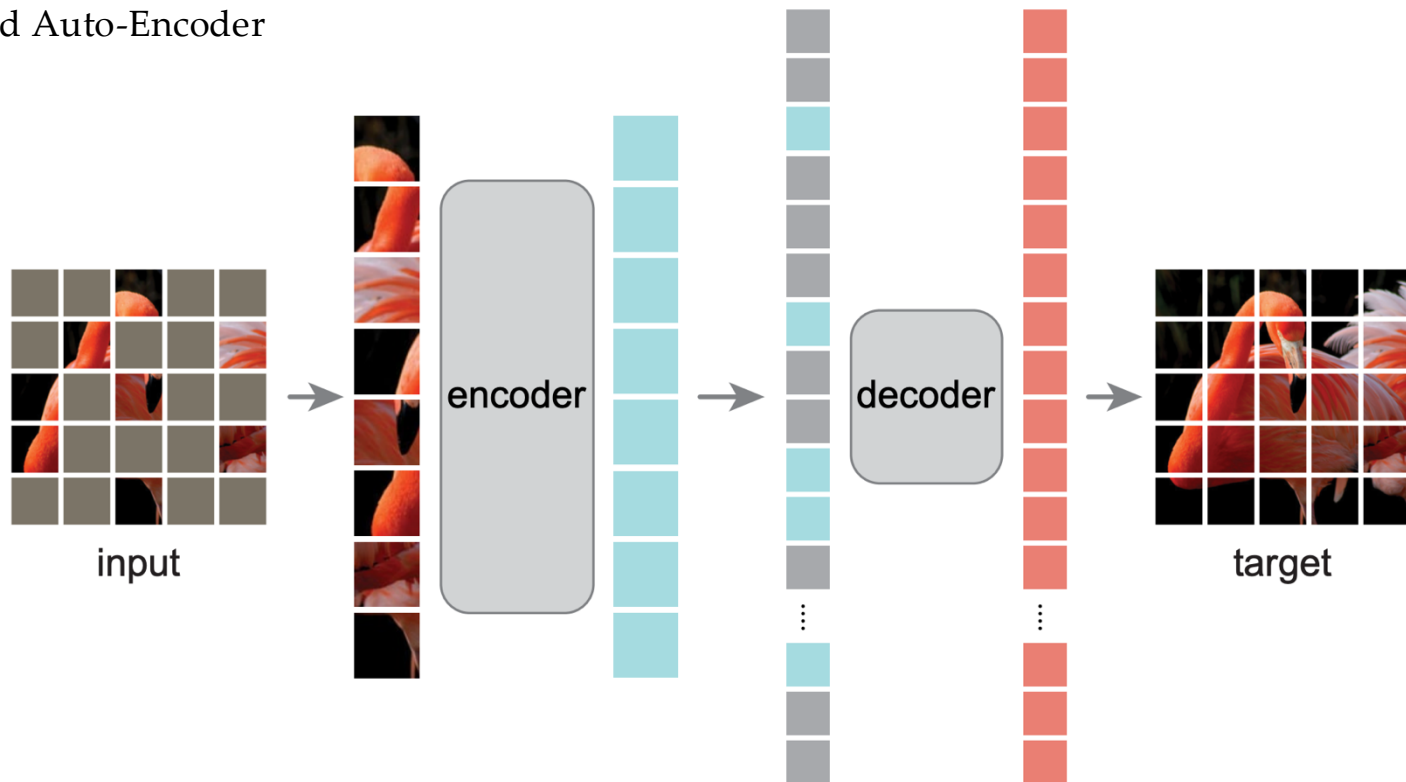
<https://ai.facebook.com/blog/dino-paws-computer-vision-with-self-supervised-transformers-and-10x-more-efficient-training>

BERT Pretraining – The Masking Approach

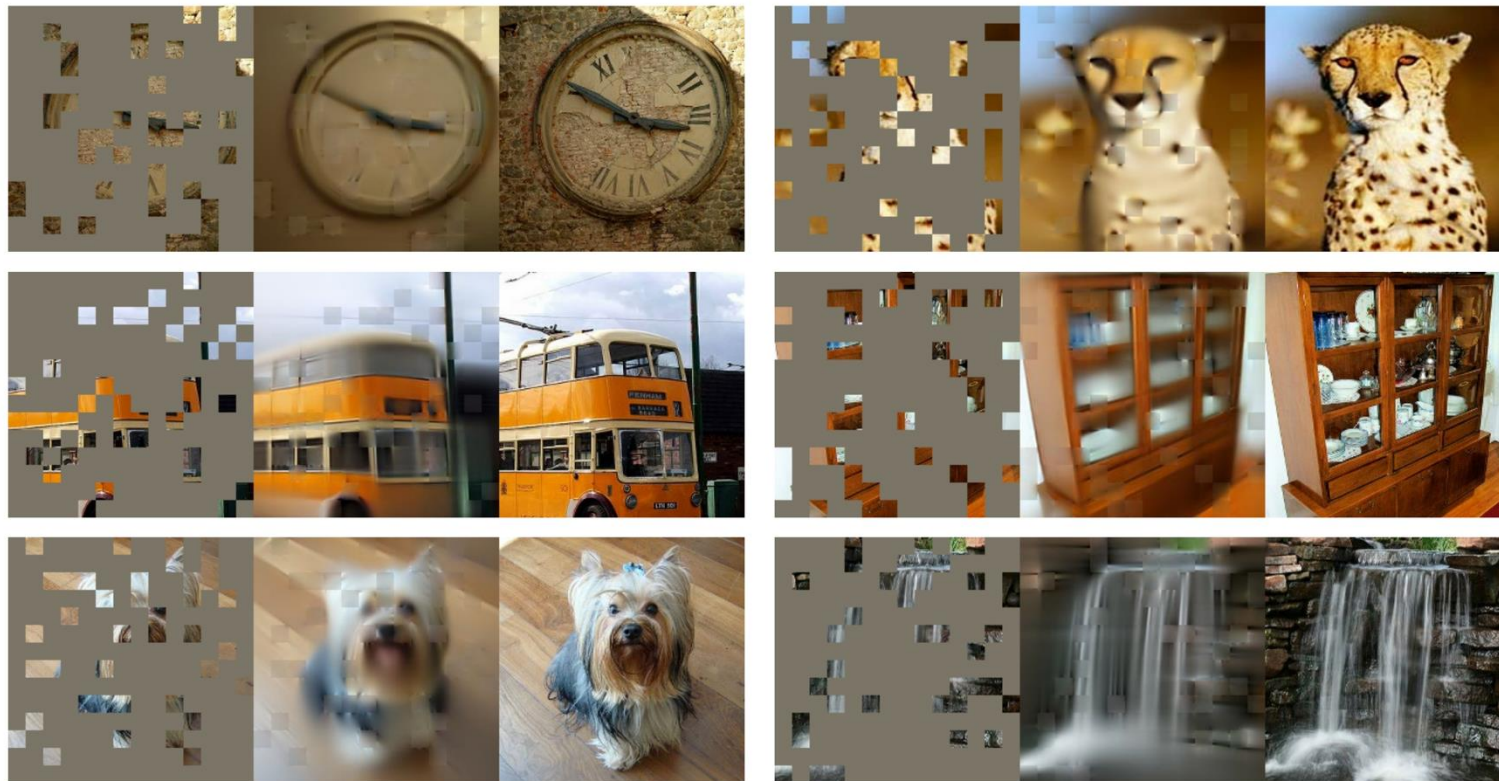


Latest Self-Supervised Learning

Masked Auto-Encoder



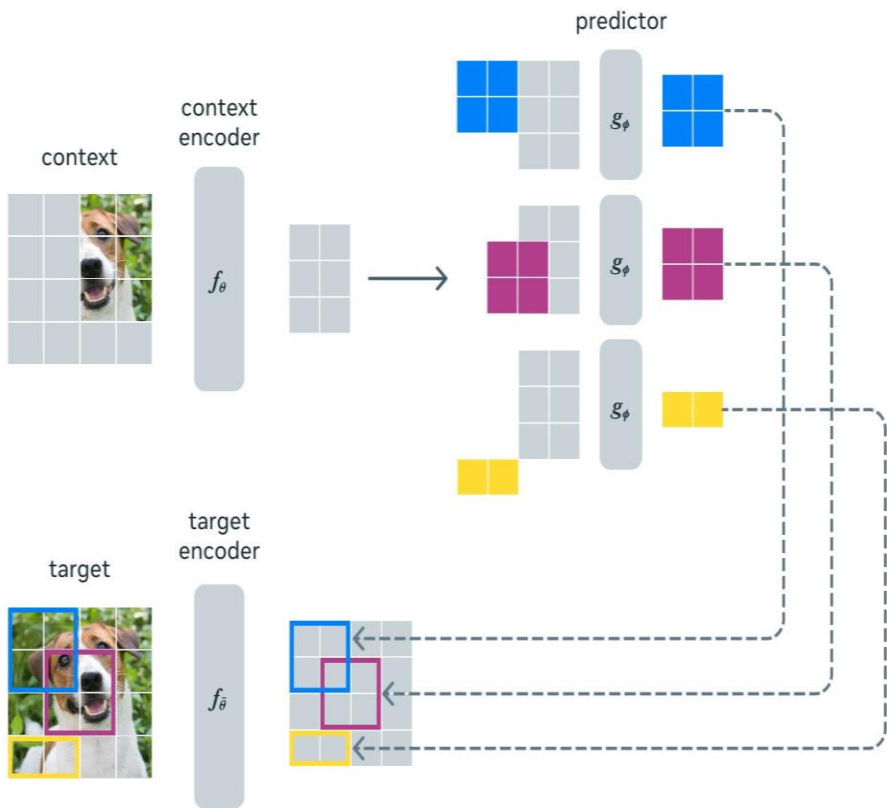
Latest Self-Supervised Learning



Joint embedding predictive architecture (JEPA)

- Break the image into patches
- Target encoder encodes each patch ($E_1, E_2, \dots, E_{256}$)
- Sample a target portion by bounding box coordinates
 - The embeddings within that region become ground truth target embeddings
- Context encoder gets the image with this target portion masked, but can see the rest
 - Processes the visible pieces and outputs their embeddings (it is ViT so has spatial knowledge)
- Predictor is a separate, lightweight network that sits between the two encoders.
 - Takes as input the context encoder's embeddings
 - Blank positions tokens specifying the location of the masked target box
 - **Predicts in embedding space the exact spatial vectors that a full-view network would generate at the missing coordinates**
- VJEPA is more suitable for 3D medical imaging
 - **V-JEPA**, a video clip is sliced into 3D **spacetime cubes**
 - Mask is over space and time or in 3D coordinates

JEPA and VJEPA



V-JEPA 2

